

Code Alpha Internship – Task 3

Secure Coding Review

Submitted By: Meeqat Arif

Domain: Cyber Security

Task: Secure Coding Review

Programming Language: C++

Repository: [MeeqatArif/CodeAlpha-Secure-Coding-Review: Secure Coding Review project for the Code Alpha Cyber Security Internship.](#)

1. Project Overview

This project reviews the security of a simple login system written in C++. Two versions were created:

- * A vulnerable login system containing common security issues.
- * A secure login system implementing basic security improvements.

The objective was to identify vulnerabilities, review the source code manually, and apply secure coding practices.

2. Tools Used

Manual Code Review

The source code was inspected manually to identify security weaknesses.

Compiler

Microsoft Visual C++ Compiler (MSVC)

Development Environment

Visual Studio Code

3. Security Vulnerabilities Found

Vulnerability 1: Hardcoded Credentials

The username and password are stored directly inside the source code.

Risk

- * Anyone with access to the code can view the credentials.

Recommendation

- * Store credentials securely using hashing or a database.

Vulnerability 2: Plain Text Password

Passwords are compared directly as plain text.

Risk

- * Passwords can be exposed.

Recommendation

- * Store hashed passwords instead of plain text.

Vulnerability 3: Unlimited Login Attempts

The vulnerable program allows unlimited login attempts.

Risk

- * Brute-force attacks become possible.

Recommendation

- * Limit login attempts.

Vulnerability 4: No Account Lock

The vulnerable program never locks the account.

Risk

- * Attackers can continue guessing passwords.

Recommendation

- * Lock the account after several failed attempts.

4. Improvements Implemented

The secure version includes:

- * Login attempt limit
- * Account lock after three failed attempts
- * Cleaner program structure
- * Better security practices

5. Manual Inspection

The source code was manually reviewed to identify:

- * Hardcoded credentials
- * Password handling
- * Login logic
- * Missing attempt limits
- * Missing account lock mechanism

No automated static analysis tool was used.

6. Program Testing

The following tests were performed.

Test 1

Correct Username: admin

Correct Password: admin123

Result

Login Successful

Test 2

Incorrect Username

Incorrect Password

Result

Invalid Username or Password

Test 3

Three incorrect login attempts

Result

Account Locked

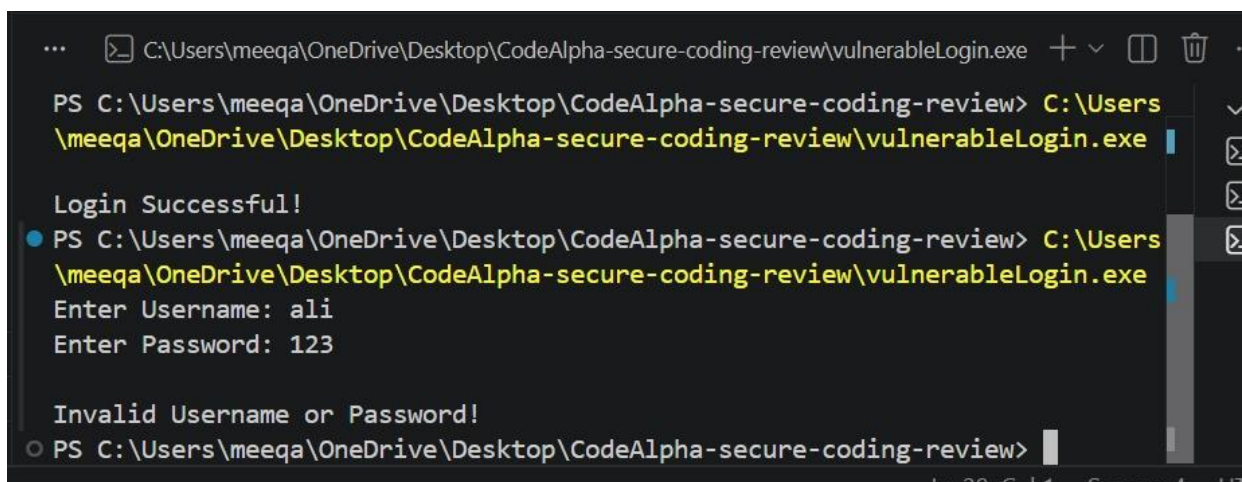
7. Conclusion

The vulnerable login system contained several common security weaknesses. After reviewing the code and implementing secure coding practices, the secure version became more resistant to brute-force attacks and improved the overall security of the login process.



```
PS C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review> C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review\vulnerableLogin.exe
Enter Username: admin
Enter Password: admin123

Login Successful!
PS C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review>
```



```
PS C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review> C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review\vulnerableLogin.exe
Login Successful!
PS C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review> C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review\vulnerableLogin.exe
Enter Username: ali
Enter Password: 123

Invalid Username or Password!
PS C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review>
```

Invalid Username or Password!

Remaining Attempts: 1

Enter Username: test

Enter Password: 000

Account Locked! Too many failed login attempts.

PS C:\Users\meeqa\OneDrive\Desktop\CodeAlpha-secure-coding-review>

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